

Uncertainty and purpose of measurements of carbon sequestration in soil

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ABSTRACT ORIGINAL

Many philosophers of measurement agree that measurements and measurement uncertainty need to be evaluated by their adequacy for a particular purpose. This paper uses an analysis of measurements of carbon in soil, which are used to support carbon trading and carbon offsetting, to test the adequacy-for-purpose method of measurement evaluation. In particular, evaluating the soil carbon case requires not just matching measurement practices to aims, but also being able to evaluate the aims of carbon trading or offsetting and being able to tell whether the measurement practices are (in)adequate for those more-ultimate aims. © 2026 Elsevier Ltd